Protect the Business Path with Zero Trust Packet Routing

OUT1015 Self-Paced Lab: Activity Guide

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Introduction

Overview

At Oracle AI World you watched this as an instructor demonstration. This guide lets you build the same environment in your own OCI tenancy and configure Zero Trust Packet Routing (ZPR) yourself.

Summit Retail's customers browse products through a public load balancer. A private application in a front-end VCN queries a private database in a back-end VCN, connected through a Dynamic Routing Gateway (DRG). To let the application reach the database, the database's network security group (NSG) admits the whole front-end VCN on the database port. That rule also admits an unrelated Test VM. Asha wants to keep the customer journey working, keep the application talking to the database, and stop the Test VM from reaching the database port.

ZPR does this with labels instead of addresses. You put security attributes on the resources, and a ZPR policy says which labelled resources may talk to each other. Traffic is allowed only when both the network rules and the ZPR policy allow it.

Estimated time: about 1 hour 15 minutes: 25 minutes of setup (mostly waiting), 35 minutes of practices, and 15 minutes of clean-up.

Architecture

Resource	Address	Purpose
Public load balancer `aiw-zpr-lb`	Public IP	Customer entry point, HTTP port 80.
Application VM `aiw-zpr-app`	10.96.10.20	Private. Queries the database for each customer request.
Test VM `aiw-zpr-test`	10.96.0.10, public IP	A troubleshooting machine with no business need for the database. You reach it with SSH.
Database VM `aiw-zpr-db`	10.97.10.20	Private. Oracle Database Free on port 1521, in the back-end VCN.
Front-end VCN 10.96.0.0/16, back-end VCN 10.97.0.0/16	-	Connected through the DRG `aiw-zpr-drg`.

Objectives

- See that an address-based NSG rule allows more than the business needs.
- Write a ZPR policy that describes the business path in terms of labelled resources.

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- Assign security attributes in enforce mode, and prove that the application still works while the Test VM is blocked.

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Get Started

Prerequisites

- An OCI tenancy in which you are an administrator.
- ZPR enabled in the tenancy, or permission to enable it. Enabling ZPR is a tenancy-wide setting; the build can do it for you with `--enable-zpr`.
- Service limits for 3 small x86 VMs (1 OCPU each), 1 flexible load balancer, 2 VCNs, and 1 DRG in your region. The VMs and load balancer are charged by the hour until you clean up.

How to Read This Guide

You see	It means
Bold text	Text you click or select in the Console.
`Code text`	A value you type, or a name you look for.
A dark box	Commands to run in Cloud Shell. Lines after the box show the expected output.
Expected result	What to check before you move on.

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Practice 1-1: Build the Starting State

One package builds both VCNs, the DRG, the three VMs, the load balancer, and the ZPR namespace with its two attributes. It also creates an administration ZPR policy that keeps SSH and software downloads working once you turn on enforcement. It also labels the two VCNs `frontend` and `backend`. A VCN label on its own changes nothing: ZPR enforces only on labelled endpoints, such as VMs and load balancers. It does not create the business policy or label any endpoint; you do that in Practices 1-2 to 1-8.

Estimated time: 25 minutes, most of it waiting for the database to install.

Download the Lab Package

1. Sign in to the OCI Console, and select the region where you want the lab. In the Console header, click the **Developer tools** icon, and select **Cloud Shell**.

2. Download and extract the package:

```
$ REPO=https://github.com/SaurabhOCI/aiworld-2026-zero-trust-labs
$ ZIP=aiw-zpr-lab.zip
$ wget -q -O ~/$ZIP $REPO/raw/main/self-paced/$ZIP
$ unzip -o -q ~/$ZIP -d ~/aiw-zpr-lab && cd ~/aiw-zpr-lab
$ ls
```

```
README.md  app.py  bootstrap-app.sh  bootstrap-db.sh
lab.py  main.tf
```

Build the Environment

1. Run the build. Add `--enable-zpr` only if ZPR is not yet enabled in your tenancy:

```
$ python3 lab.py build
```

```
Prepared: region ..., VM.Standard.E5.Flex, image ...,
  namespace aiwzpr...
Creating the lab compartment
Creating the ZPR namespace and attributes
Creating the lab resources (Terraform apply, about 5 minutes)
Load balancer 203.0.113.25. Waiting for the database VM ...
Customer query works: the application reached the database.
...
ZPR lab: STARTING STATE
```

2. Write down two values from the output: the load balancer address, and your namespace, which starts with `aiwzpr`. You use both in Practices 1-2 to 1-8.

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> **Note**
>
> To create the lab under another compartment, add `--parent <compartment-OCID>`.
>
> If the build stops with a message, fix what it says and run the same command again. It continues where it stopped.
>
> The build may print `Waiting 60 seconds for OCI to distribute the new ZPR attributes`. That is normal: new labels take a few minutes to reach every region.
>
> Your namespace has a random suffix, for example `aiwzpr4k2m9x`. In this guide it is written as `<ns>`. Replace it everywhere.

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Practice 1-2: Use the Business Path

Keep one picture in mind: customer to load balancer, load balancer to application, application to database. The Test VM sits in the same front-end VCN as the application.

Estimated time for Practices 1-2 to 1-8: 35 minutes.

1. Open `http://<load-balancer-address>/` in your browser. The Summit Retail page shows three products from the database.

2. In Cloud Shell, check the same query:

```
$ cd ~/aiw-zpr-lab
$ curl -s http://<load-balancer-address>/dbcheck
```

```
{"status": "DB connection successful", "products": [...]}
```

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Practice 1-3: Find What the Network Allows

1. From the navigation menu, select **Networking** > **Virtual cloud networks**. Select the compartment `aiw-zpr-lab`, and click `aiw-zpr-backend-vcn`.

2. Click the **Security** tab, and then **Network security groups**. Click `aiw-zpr-db-nsg`, and look at the ingress rule on TCP port 1521.

   > **Expected result**
   >
   > The rule admits `10.96.0.0/16`, the whole front-end VCN. The application (10.96.10.20) needs it, but the Test VM (10.96.0.10) is in the same range.

3. Ask the Test VM to connect to the database port:

   ```
   $ python3 lab.py check
   ```

   ```
   DATABASE PORT REACHABLE
   ```

   > **Note**
   >
   > The check opens a TCP connection to port 1521 and nothing more. It does not sign in to the database or read data.

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Practice 1-4: State the Intent

Before you write any policy, say what should happen, in business terms:

- Customers use the website on port 80.
- The load balancer forwards customers to the application on port 80.
- The application queries the database on port 1521, across the two VCNs.
- Nothing else may reach the database port, including the Test VM.

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Practice 1-5: Look at the Prepared Labels

1. Switch the Console to your **home region**. From the navigation menu, select **Identity & Security** > **Zero Trust Packet Routing**, and then **Security attribute namespaces**. Select the compartment `aiw-zpr-lab`, and open `<ns>`.

2. Look at its two attributes and their allowed values:

Attribute	Allowed values	You assign it to
`networks`	`frontend`, `backend`	Already assigned to the two VCNs by the build
`role`	`loadbalancer`, `application`, `database`, `test`	The load balancer and the three VMs

3. Switch to the lab region, open **Networking** > **Virtual cloud networks**, and check the **Security** tab of `aiw-zpr-frontend-vcn`: it carries `networks` = `frontend`. The business policy's `in ... VCN` conditions use these labels.

> **Note**
>
> In testing, ZPR honoured the `in ... VCN` conditions only for VCNs that carried their label from creation. That is why the build labels the VCNs for you. If you label VCNs in your own designs, label them when you create them.

4. Back in the home region, open **Zero Trust Packet Routing** > **Policies**, and read `aiw-zpr-<ns>-administration`. It allows SSH to and from the Test VM and outbound web traffic for updates. It has no statement that lets the Test VM reach port 1521.

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Practice 1-6: Write the Business Policy

1. Still in the home region, in **Zero Trust Packet Routing** > **Policies**, click **Create policy**. Enter the name `aiw-zpr-business-path` and a description.

2. Select **Add policy statements** > **Manual policy builder**, and enter these three statements. Replace `<ns>` with your namespace:

```
$ in <ns>.networks:frontend VCN allow '0.0.0.0/0'
  to connect to <ns>.role:loadbalancer endpoints
  with protocol='tcp/80'
$ in <ns>.networks:frontend VCN
  allow <ns>.role:loadbalancer endpoints
  to connect to <ns>.role:application endpoints
  with protocol='tcp/80'
$ allow <ns>.role:application endpoints
  in <ns>.networks:frontend VCN
  to connect to <ns>.role:database endpoints
  with protocol='tcp/1521' in <ns>.networks:backend VCN
```

> **Note**
>
> Each statement is one line in the policy builder; it is split here to fit the page. Read each one as: who may connect to whom, on which port.

3. Click **Add**, and then **Create policy**. Wait until the policy is **Active**.

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Practice 1-7: Assign the Labels

1. Switch back to the lab region. On each of the four resources below, open its details page, find **Security attributes** (on the resource's **Security** tab), and click **Add security attributes**. Select namespace <ns>, the attribute, the value, and mode **Enforce**:

Resource	Where	Attribute	Value
aiw-zpr-lb	Networking > Load balancers	role	loadbalancer
aiw-zpr-app	Compute > Instances	role	application
aiw-zpr-db	Compute > Instances	role	database
aiw-zpr-test	Compute > Instances	role	test

2. Wait until all four show the label. Then allow a few minutes for enforcement to settle before you test.

```
$ python3 lab.py status
```

```
...
ZPR lab: PROTECTED
```

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Practice 1-8: Prove the Intent

1. Run the same two tests as before:

```
$ curl -s http://<load-balancer-address>/dbcheck
$ python3 lab.py check
```

```
{"status": "DB connection successful", "products": [...]}
DATABASE PORT BLOCKED: timeout
```

> **Expected result**
>
> The customer query still works: the application reached the database on the path the policy allows.
>
> The Test VM's connection to port 1521 times out. Its address, the routes, and the NSG rule are unchanged; it no longer matches an allowed relationship.

> **Note**
>
> An SSH error or a refused connection is not the ZPR result. Only `timeout` shows that ZPR dropped the traffic.

Checkpoint: What You Proved

Connection	Before ZPR	After ZPR
Customer to website (port 80)	Works	Works
Application to database (port 1521)	Works	Works
Test VM to database (port 1521)	Reachable	Timeout

- Both VMs are still in the front-end VCN, and the NSG rule still admits the whole VCN. Why can only the application reach port 1521?
- Does this prove that the Test VM cannot sign in to the database?

> **Checkpoint**
>
> **1.** The application's label matches the policy's application-to-database statement. The Test VM's label `test` matches no statement on port 1521, and with enforce mode, unmatched traffic is dropped.
>
> **2.** No. The test proves only that the connection is blocked. Database accounts and passwords remain a separate control.

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Repeat the Practices

To start again, reset the lab. It removes your four labels and deletes any ZPR policy that uses your namespace, except the administration policy:

```
$ cd ~/aiw-zpr-lab
$ python3 lab.py reset
```

```
Removed the <ns> label from aiw-zpr-app
...
ZPR lab: STARTING STATE
```

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Troubleshooting

- `Zero Trust Packet Routing is not enabled`: run the build with `--enable-zpr`, or enable it in **Identity & Security** > **Zero Trust Packet Routing**.
- `No suitable x86 flexible shape`, or `LimitExceeded` (for example `Limit for DRGs per tenant`): the region has no room for the lab. Run `python3 lab.py teardown --yes`, then build in another region with `--region <region>`, for example `--region us-chicago-1`. Because the old compartment name stays reserved for a while, also add `--parent <compartment-OCID>` with a different parent compartment.
- To avoid the DRG limit, check before you build: the build uses your Cloud Shell's region unless you add `--region`. The lab needs one free dynamic routing gateway (DRG) in that region. Compare the DRGs in **Networking** > **Dynamic routing gateways** with the limit in **Governance & Administration** > **Limits, Quotas and Usage** (service **Networking**, `DRG count`).
- The customer query stopped working after Practice 1-7: wait a few minutes and try again. Then check that all four resources carry the label with mode **Enforce**, and that all three business statements use your namespace.
- `python3 lab.py check` reports an SSH error: wait a minute and try again. The check needs the Test VM running and your Cloud Shell able to reach its public address.
- Anything else: `python3 lab.py status` shows every label, the policies, and both tests.

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Practice 1-9: Clean Up Your Tenancy

Remove everything the lab created. The teardown first removes your labels and business policy, then deletes the resources:

```
$ cd ~/aiw-zpr-lab
$ python3 lab.py teardown --yes
```

```
...
Deleting compartment aiw-zpr-lab (30 minutes or more)
Teardown complete.
OCI finishes deleting the compartment in the background.
```

> **Note**
>
> OCI keeps the name `aiw-zpr-lab` reserved until the compartment is deleted. To build the lab again sooner, use `--parent` to build it in a different compartment.
>
> The teardown can sit on `Still destroying` for a security attribute for 10 to 20 minutes. Security attributes are defined for the whole tenancy and copied to every subscribed region, and OCI deletes them in the background after they are retired. Let it run; if Terraform stops with a timeout, run the same teardown again.
>
> Teardown leaves ZPR enabled for the tenancy. If you enabled it only for this lab and want it off, disable it in **Zero Trust Packet Routing** after the teardown.

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Acknowledgements

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